

needed to regularly monitor and transmit information related to the population, water quality, etc.

Low-cost cardiac diagnostic system

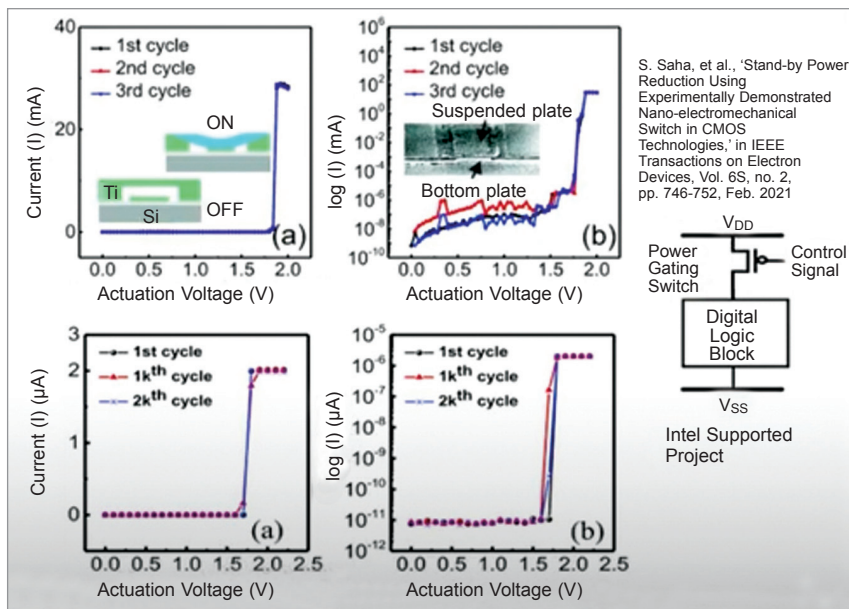
Back in 2004-05, we formed a multidisciplinary group and started to think about what all could be improved and address the requirements at the same time. One of the first problems that we took on was a low-cost cardiac diagnostic system. Every third death in India is cardiac related. We strived to make the diagnosis simpler for everyone at the lowest cost possible.

There are multiple cardiac markers to open in CK MB myoglobin markers. We started working on some of these markers, because sometimes the cardiac muscles release some of these proteins into the blood when a patient starts to die. If troponin concentration is high, one needs to be admitted to a hospital immediately. With the help of some simple technologies these markers can be detected. But these markers need to be tested repetitively as it is not a one-time measurement need.

We wanted to build such a system, so we started working on iSens that could detect myocardial infarction. You just need to take the blood sample and filter it out. There are micro channels there along with the biosensor array for different cardiac markers and then there is a small tabletop system.

At the time of development, we realised that these are all disposable systems and silicon devices turned out to be pretty expensive to be disposed of frequently. While all the associated electronics are made of silicon but for the actual cartridge we started making use of polymer sensors. These are all piezo-resistive polymer cantilevers that can be implemented with low-cost techniques.

When an antibody gets coated on the cantilever and binds to the



A cycle of the nano mechanical switch

cardiac proteins, it causes a reaction and generates a stress, and that stress goes on to cause a strain. Any binding of the cardiac protein to the cantilever surface gives rise to an electrical signal, and that is how you know whether the cardiac markers are of a high or low category.

A new approach to explosive detection

Using the same technology, we started working on explosives, because in India security is a major concern. We wanted to develop a simple system that could be used to screen people at the airports and even in buses and trains. We stumbled upon a very interesting technology for which we have been granted patents in the United States and India.

Why we need to miniaturise the diagnostic systems

- Low sample volume
- Faster reaction - lesser diffusion length for analyte-receptor interactions and lower cost of fabrication
- Multi-analyte detection
- Lower cost of consumables (antibodies)
- Higher surface to volume ratio - higher sensitivity

In this technology we make use of a nano mechanical switch between two plates. The distance between the two plates is approximately 20 nanometers. This distance becomes narrower when explosives bind on the surface of one of the plates. When this gap becomes 5 to 10 nanometres between the plates, a rush of current occurs. This creates a lot of thermal energy, which leads to a thermal shock and thus to an explosion. That nano explosion destroys the switch.

We started fabricating these switches in such a way that they could withstand multiple cycles (see figure above). The switches can now close and open up to 2000 times.

Even Intel was very keen on using this technology for standby platforms. So, we started a company called Nanosniffer using this technology. Last year it was released as a product. This technology is now commercially being used at airports. It is important to know that it takes about ten seconds only to do the detection.

All the explosive detectors at airports and other places were earlier based on spectroscopy principles, which made them very expensive.